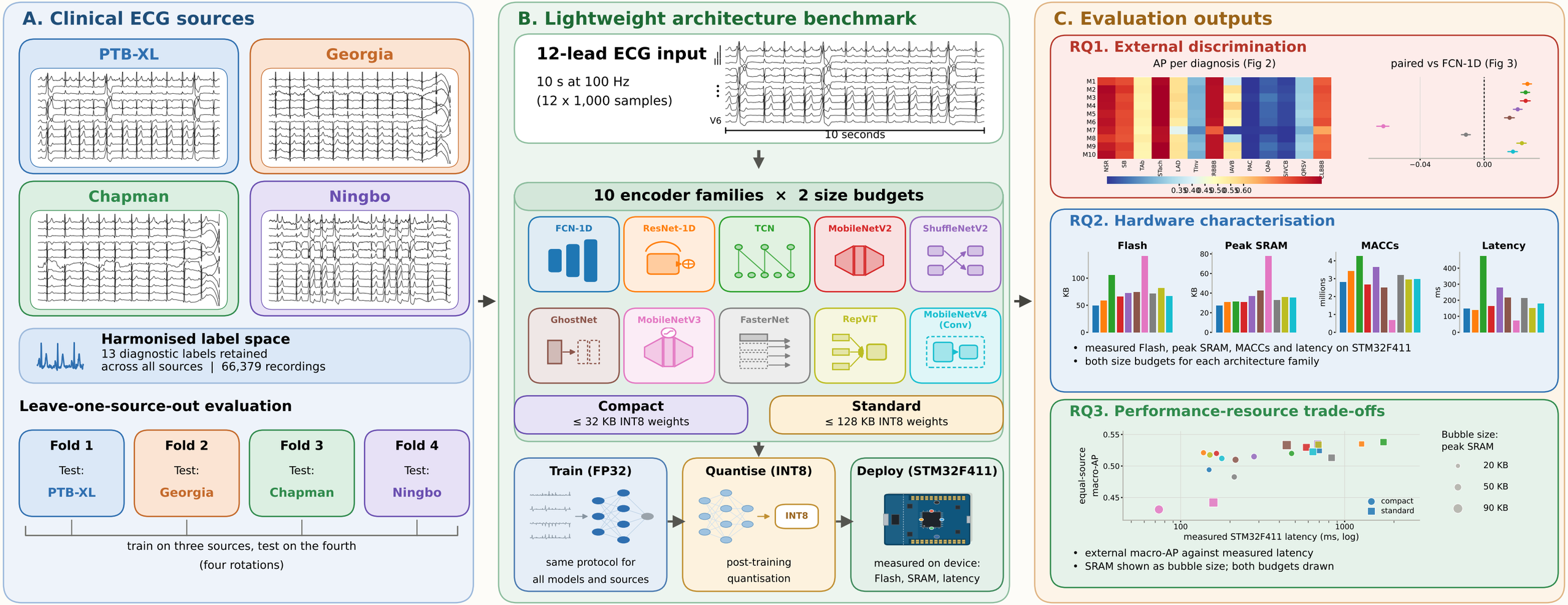

A. Clinical ECG sources
PTB-XL
Georgia
Chapman
Ningbo
Harmonised label space
13 diagnostic labels retained
across all sources | 66,379 recordings
Leave-one-source-out evaluation
Fold 1
Test:
PTB-XL
Fold 2
Test:
Georgia
Fold 3
Test:
Chapman
Fold 4
Test:
Ningbo
train on three sources, test on the fourth
(four rotations)
B. Lightweight architecture benchmark
12-lead ECG input
10 s at 100 Hz
(12 x 1,000 samples)
V6
10 seconds
10 encoder families × 2 size budgets
FCN-1D
ResNet-1D
TCN
MobileNetV2
ShuffleNetV2
GhostNet
MobileNetV3
FasterNet
RepViT
MobileNetV4 (Conv)
Compact
≤ 32 KB INT8 weights
Standard
≤ 128 KB INT8 weights
Train (FP32)
same protocol for
all models and sources
Quantise (INT8)
INT8
post-training
quantisation
Deploy (STM32F411)
measured on device:
Flash, SRAM, latency
C. Evaluation outputs
RQ1. External discrimination
AP per diagnosis (Fig 2)
M1
M2
M3
M4
M5
M6
M7
M8
M9
M10
NSR
SB
TAb
STach
LAD
TInv
RBBB
IAVB
PAC
QAb
SVCB
QRSV
LBBB
0.35
0.40
0.45
0.50
0.55
0.60
paired vs FCN-1D (Fig 3)
−0.04
0.00
RQ2. Hardware characterisation
Flash
Peak SRAM
MACCs
Latency
KB
millions
ms
• measured Flash, peak SRAM, MACCs and latency on STM32F411
• both size budgets for each architecture family
RQ3. Performance-resource trade-offs
equal-source macro-AP
measured STM32F411 latency (ms, log)
compact
standard
Bubble size:
peak SRAM
20 KB
50 KB
90 KB
• external macro-AP against measured latency
• SRAM shown as bubble size; both budgets drawn